

Lie Alg: $\mathfrak{g}$ is a vec. space w/ binary op.

$[\cdot, \cdot]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ w/ the Jacobi:

Rep: $\rho: \mathfrak{g} \rightarrow \text{Aut}(V)$, Homomorphism wrt. $[\cdot, \cdot]$.

$R \equiv \dim(V)$

Generators: $\{t^a\}$ $[t^a, t^b] = f^{ab}_c t^c$

* Notes use $\tilde{t} \equiv i t$.

Simple Lie Alg: is a LA that contains no proper ideals (no proper subset of gens $\{t^a\}$ s.t. $[L^a, t^a] \in \{t^a\} \forall t^a$).

Semi-simple: Direct sum of Simple

Cartan-Weyl Basis:

• May want to work independent of a specific basis. Rep theory: specific basis is wanted
 $\rightarrow$ Create a 'canonical' basis.

Rank: The Cardinality of the set of of
 Maximal commuting ^{Hermitian} Generators H_i

$$\left\{ \begin{array}{l} |\{H_i\}|_{i=1}^r = r \equiv \text{Rank}(\mathfrak{g}) \\ \text{s.t. } [H_i, H_j] = 0 \end{array} \right.$$

$\{H_i\}$: This set is the Cartan Subalgebra

$$\mathfrak{h} \subset \mathfrak{g}, \quad \mathfrak{h} \equiv \{H^i \in \text{Span}(t^i) \mid [H^i, H^j] = 0\}$$

* The generators H^i can all be simult. diag. since they are Hermitian.

$$\text{i.e. } \exists Q \in \mathfrak{so}(r) \text{ s.t. } Q^T H^i Q = \text{diag}(\cdot)$$

$$\Delta = \mathfrak{h} \oplus (\Delta \setminus \mathfrak{h})$$

$$(\Delta \setminus \mathfrak{h}) = \text{Span}(E^\alpha)$$

We choose E^α s.t.

$$[H^i, E^\alpha] = \alpha^i E^\alpha$$

$\hookrightarrow$ i.e. $(\Delta \setminus \mathfrak{h})$ is an ideal of Δ

$$\alpha \equiv (\alpha^1, \dots, \alpha^r) : \text{Root}$$

α is called a root as it is a root of the characteristic eq for H^i

$$Q^T H^i Q = \text{diag}(\alpha^1, \dots, \alpha^r)$$

$$E^\alpha : \text{Ladder operators}$$

Def: A basis of $\Delta = \text{Span}\{H^i\}^r \oplus \text{Span}\{E^\alpha\}$ which satisfies $[H^i, H^j] = 0$ & $[H^i, E^\alpha] = \alpha^i E^\alpha$ is called a Cartan-Weyl Basis

$\mathfrak{h}$: "Maximal Abelian Subalgebra of Δ "

The roots are non-degenerate, i.e. unique

$$[H^i, H^j] = 0 \text{ or } H^i \in \mathfrak{H}^0 - \dots$$

$$= \text{diag}(\alpha_1^i, \dots, \alpha_n^i) - \text{diag}(\alpha_1^j, \dots, \alpha_n^j) = 0$$

* The roots can be seen as dual elements of $\mathfrak{h}$

$$\text{i.e. } \mathfrak{K}^* = \{ \alpha \mid \alpha \cdot \mathfrak{h} \rightarrow \mathbb{R} \}$$

$$\alpha(H^i) = \alpha^k \hat{\otimes}^k(H^i) = \alpha^k \delta_k^i = \alpha^i$$

$$\boxed{\alpha(H^i) = \alpha^i}$$

$$\text{Note: } ([H^i, E^\alpha])^\dagger = \alpha^i (E^\alpha)^\dagger$$

$$= [(E^\alpha)^\dagger, H^i] = \alpha^i (E^\alpha)^\dagger$$

$$\Rightarrow \boxed{[H^i, (E^\alpha)^\dagger] = (-\alpha^i) (E^\alpha)^\dagger}$$

Thus, $(-\alpha^i)$ is also a root with

$$\boxed{(E^\alpha)^\dagger = E^{-\alpha}}$$

* We define the set of roots $\mathfrak{K}^*$

$$\boxed{\Delta \subset \mathfrak{K}^* = \{\text{set of all roots}\}}$$

$\rightarrow$ Called the Root system.

* α^i are the Eigenvalues of H^i in the adjoint rep.

$$\underline{\text{ad}(X): \mathfrak{g} \rightarrow \mathfrak{g}}, \quad \text{ad}(X) =: \text{ad}_X$$

$$\boxed{\text{ad}_X(Y) =: [X, Y]}$$

$$\text{Note } \text{ad}: \mathfrak{g} \rightarrow \text{Aut}(\mathfrak{g})$$

$$\text{but here } \mathfrak{g} = \mathfrak{g}, \text{ so } \text{ad}: \mathfrak{g} \rightarrow \text{Ad}(\mathfrak{g})$$

Matrix rep of adjoint:

$$(t^a)_{bc} = -f_{abc}$$

$$E^\alpha \mapsto |E^\alpha\rangle \equiv |\alpha\rangle$$

$$H^i \mapsto |H^i\rangle$$

$$\text{ad}_{H^i} E^\alpha = \alpha^i E^\alpha \Leftrightarrow \hat{H}^i |\alpha\rangle = \alpha^i |\alpha\rangle$$

$$H^i \equiv \text{ad}_{H^i} : \mathfrak{g} \rightarrow \mathfrak{g}$$

- 1-to-1 correspondence between $|\alpha\rangle$ and E^α reflect the fact that α are non degenerate

* In the adjoint rep, the zero eigen value has degeneracy $\leq (\text{acting on } H^i)$ on all of $\mathfrak{g}$ since $H^i |H^j\rangle = 0 \quad \forall \quad i, j = 1, \dots, r$

$$\dim(\mathfrak{h}^c) + (\dim(\Delta))$$

$$\dim(\text{adj. rep}) = \dim(\mathfrak{g}) = r + \# \text{ roots}$$

* From the Jacobi - identity

$$0 = [H^i, [E^\alpha, E^\beta]] + [E^\alpha, [E^\beta, H^i]] + [E^\beta, [H^i, E^\alpha]]$$

$$[H^i, [E^\alpha, E^\beta]] = [E^\alpha, \alpha^i E^\beta] + [E^\beta, \beta^i E^\alpha]$$

$$[H^i, [E^\alpha, E^\beta]] = (\alpha^i + \beta^i) [E^\alpha, E^\beta]$$

$$\begin{cases} [E^\alpha, E^\beta] \propto E^{\alpha+\beta} & \text{if } \alpha+\beta \in \Delta \\ [E^\alpha, E^\beta] = 0 & \text{if } \alpha+\beta \notin \Delta \\ [E^\alpha, E^\beta] \text{ is a linear comb. of } H^i & \alpha = -\beta \end{cases}$$

* Note, Δ , $\mathfrak{h}$, $\mathfrak{k}$ all have linear geometries and we can define $\mathbb{R}$ inner products

$$\alpha \cdot H \equiv \alpha(H) = \alpha_i H^i, \quad i \in \{1, \dots, r\}$$

$$|\alpha|^2 = \sum_{i=1}^r \alpha_i^2$$

$$\begin{aligned} [H^i, H^j] &= 0, \quad [E^\alpha, E^\beta] = \begin{cases} N_{\alpha, \beta} E^{\alpha+\beta}, & \text{if } \alpha+\beta \in \Delta \\ \frac{2}{|\alpha|^2} \alpha \cdot H & \alpha = -\beta \\ 0 & \text{otherwise} \end{cases} \\ [H^i, E^\alpha] &= \alpha^i E^\alpha \\ N_{\alpha, \beta} &: \text{constant} \end{aligned}$$

$$\bullet [E^\alpha, E^{-\alpha}], \quad \frac{2}{|\alpha|^2} \alpha \cdot H \quad [E^\alpha, E^{-\alpha}] = 2b^i H^i$$

$$[H^i, \sum b^i H^i] = 0$$

So, for $[E^\alpha, E^\beta]$ we have simply normalized the constants in front of H w/ a factor of 2.

ex: Cartan Weyl basis of $sl(2, \mathbb{C})$:

• We want to find the maximal subgroup

$$\text{of } \mathcal{SL}(2, \mathbb{C}). \begin{pmatrix} a & b \\ c & d \end{pmatrix}, ad - bc = 1$$

$$\frac{d}{dt} e^{tA} \Big|_{t=0} = A e^{tA} \Big|_{t=0} = A$$

$$\det(e^{At}) = 1 + t \operatorname{tr}(A) = 1$$

$$\Rightarrow \boxed{\operatorname{tr}(A) = 0}$$

$$\text{So } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \rightarrow a + d = 0 \Rightarrow d = -a$$

$$A = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$$

$$\bullet \text{ Let } B \in \mathcal{SL}(2, \mathbb{C}) : B = \begin{pmatrix} \tilde{a} & \tilde{b} \\ \tilde{c} & -\tilde{a} \end{pmatrix}$$

$$AB = BA$$

$$\begin{pmatrix} a & b \\ c & -a \end{pmatrix} \begin{pmatrix} \tilde{a} & \tilde{b} \\ \tilde{c} & -\tilde{a} \end{pmatrix} = \begin{pmatrix} a\tilde{a} - \tilde{c}b & \tilde{b}a - \tilde{a}b \\ \tilde{a}c - a\tilde{c} & \tilde{b}c - \tilde{a}a \end{pmatrix}$$

$$\stackrel{!}{=} \begin{pmatrix} a\tilde{a} - c\tilde{b} & b\tilde{a} - a\tilde{b} \\ a\tilde{c} - \tilde{a}c & b\tilde{c} - a\tilde{a} \end{pmatrix}$$

$$\Rightarrow \tilde{c}b = c\tilde{b}$$

$$\tilde{b}a - \tilde{a}b = b\tilde{a} - a\tilde{b}$$

$$a\tilde{c} - \tilde{a}c = a\tilde{c} - a\tilde{c}$$

$$\leadsto \text{Take } a=1, b=c=0$$

$$\Rightarrow H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$\therefore$

$$\left. \begin{array}{l} \tilde{b} = -\tilde{b} \\ \tilde{c} = -\tilde{c} \end{array} \right\} \tilde{b}, \tilde{c} \stackrel{!}{=} 0$$

$$\text{So } B = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & -d \end{pmatrix} \stackrel{!}{=} \begin{pmatrix} a & 0 \\ 0 & +d \end{pmatrix}$$

$$\Rightarrow d=0$$

$$\boxed{E = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}$$

$$\boxed{H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}} \quad \boxed{F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}}$$

NOTE

$$HF = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} = -F$$

$$[H, E] = 2E, [H, F] = -2F, [E, F] = H$$

The Killing Form:

- To define a scalar product on $\mathfrak{g}$ we define the Killing form

$$\kappa(X, Y) := \text{Tr}(\text{ad}(X) \circ \text{ad}(Y)), \quad X, Y \in \mathfrak{g}$$

- $\kappa(Y, X) = \kappa(X, Y)$ (Symmetric)
- $\kappa([X, Y], Z) = \kappa(X, [Y, Z])$ (Invariance)

$\rightarrow$ For semi-simple LA's the Killing form is non-degenerate: $\Leftrightarrow$

$$\boxed{\kappa(X, Y) = 0, \forall Y \Leftrightarrow X = 0}$$

Killing form of $sl(2, \mathbb{C})$:

- Need $\text{ad}(E), \text{ad}(H), \text{ad}(F)$

$$\text{ad}(H)(E) = [H, E] = 2E$$

$$\Rightarrow \text{ad}(H)E = 2E$$

- For some reason, the notes embed these matrices into $M_{3 \times 3}(\mathbb{C})$

$$\text{if } \text{ad}(H)E = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \stackrel{!}{=} 2 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\Rightarrow \text{ad}(H) = 2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

however, the $\dim(\mathfrak{sl}(2, \mathbb{C})) = \underline{3}$

So the adjoint $(f^a)_{bc}$ goes from $a, b, c = 1, 2, 3$

$$\underline{\text{SO}} \quad (f^a)_{bc} \in M_{3 \times 3}(\mathbb{C})$$

we can determine $(f^a)_{bc}$ by the comm.

$$[H, E] = f_{HE}^E E = 2E$$

$$[H, H] = 0 \Rightarrow f_{HH}^a = 0 \quad (\text{as expected})$$

$$[H, F] = -2F = f_{HF}^F F$$

$$[E, F] = H = f_{EF}^H H$$

$$f_{HE}^E =$$

$$H \cong 2$$

$$E \cong 1$$

$$F \cong 1$$

$$f_{HE}^E \Rightarrow + f_{EE}^H = -2, \quad E \cong 1$$

$$(f^H)_{ab} = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Since $f_{FF}^H = 2$

$$\therefore \boxed{\text{ad}(H) = -(f^H)_{ab} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix}}$$

$$\text{ad}(E) = \begin{pmatrix} 0 & -2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \text{ad}(F) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\kappa(E, E) = 0, \quad \kappa(E, H) = 0, \quad \kappa(E, F) = 4$$

$$K(U, H) = 8, \quad K(U, F) = 0, \quad K(F, F) = 0$$

$$\text{or } K(X, Y) = \begin{pmatrix} 0 & 0 & 4 \\ 0 & 8 & 0 \\ 4 & 0 & 0 \end{pmatrix}$$

Normalized Killing form: $K(X, Y) = \frac{1}{2} g^{\mu\nu} \text{Tr}(\text{ad}_X \text{ad}_Y)$

$W^\vee$: dual cocharacter $\mathbb{H}$

• Standard basis $\{e^a\}$ is ortho curb. K

$$K(e^a, e^b) = \delta^{ab}$$

$$\hookrightarrow \text{Cartan } K(H^i, H^j) = \delta^{ij}$$

* Can use K_{ab} to raise & lower indices

$$f^a_{bc} = \sum_d f^{ad}_c [K(e^d, e^b)]^{-1}$$

* We can lower f^a_{bc} :

$$f_{abc} := (K(e^d, e^a))^{-1} f^d_{bc}$$

and so f_{abc} is totally anti-symmetric.

$$\left\{ \begin{array}{l} a \leftrightarrow b: f_{bac} = (K(e^d, e^b))^{-1} f^d_{ac} \\ \text{use Jacobi} \end{array} \right\}$$

* For an ortho-normal basis $\{T^a\}$, the position of indices is irrelevant. Note the Cartan-Weyl basis is not an ortho-basis

$$\text{Invariance } K([X, Y], Z) = \text{Tr}(\text{ad}_X \text{ad}_Y \text{ad}_Z)$$

$$\begin{aligned}
 \text{ad}[X,Y] &= [\text{ad}X, \text{ad}Y] Z \\
 \text{ad}[X,Y](Z) &= [[X,Y], Z] \\
 \text{RHS} &= [\text{ad}_X \circ \text{ad}_Y, Z] - \text{ad}_Y \circ \text{ad}_X [Z] \\
 &= [X, [Y, Z]] - [Y, [X, Z]] \\
 &= [X, [Y, Z]] + [Y, [Z, X]] \\
 &\stackrel{\text{Jacobi}}{=} [Z, [X, Y]] = [[X, Y], Z] \\
 &= \text{ad}[X,Y] Z \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \star K([X,Y], Z) &= \text{tr}(\text{ad}_{[X,Y]} \circ \text{ad}_Z) = \text{tr}([a_X a_Y] \circ a_Z) \\
 &= \text{tr}(a_X a_Y a_Z - a_Y a_X a_Z) \\
 &= \text{tr}(a_X a_Y a_Z - a_X a_Z a_Y) \\
 &= \text{tr}(a_X [a_Y, a_Z]) = \text{tr}(a_X \text{ad}[Y, Z]) \\
 &= K(X, [Y, Z]) \quad \checkmark
 \end{aligned}$$

$$\star K([Z, X], Y) + K(X, [Z, Y]) = 0$$

$\leadsto$ Note, $B_{ij} = B_{ji}$

$$\begin{aligned}
 (\text{ad}(e_i) \circ \text{ad}(e_j))(e_k) &= \text{ad}(e_i)(f_{jk}^m e_m) = f_{jk}^m f_{im}^l e_l \\
 &= (B_{ij})_k^l e_l
 \end{aligned}$$

$$\Rightarrow \text{ad}(e_i) \circ \text{ad}(e_j) = (B_{ij})_k^l$$

$$\Rightarrow \text{Tr}(B_{ij})_k^k = B_{ij})_k^k = f_{jk}^m f_{im}^k$$

$$\Rightarrow K_{ij} = f_{jk}^m f_{im}^k$$

$$[E^\alpha, E^{-\alpha}] = K(E^\alpha, E^{-\alpha}) \alpha \cdot H$$

$$K(E^\alpha, E^{-\alpha}) = 2/|\alpha|^2$$

$$\left\{ \begin{aligned} K([E^\alpha, E^{-\alpha}], [E^\alpha, E^{-\alpha}]) &= 2 K(E^\alpha, E^{-\alpha}) \\ (\text{Somehow, expand to see?}) \end{aligned} \right\}$$

Proof:

$$\begin{aligned} \tilde{K}([E^\alpha, E^{-\alpha}], [E^\alpha, E^{-\alpha}]) &= \tilde{K}(E^\alpha, [E^\alpha, [E^\alpha, E^{-\alpha}]] \\ &\quad [E^{-\alpha}, [E^\alpha, E^{-\alpha}]] + [E^\alpha, [E^{-\alpha}, E^{-\alpha}]] + [E^{-\alpha}, [E^{-\alpha}, E^\alpha]]) \\ [E^\alpha, [E^\alpha, E^{-\alpha}]] &= 0 \\ &= \tilde{K}(E^\alpha, [E^{-\alpha}, \frac{2}{|\alpha|^2} \alpha \cdot H]) \\ &= \frac{2}{|\alpha|^2} \tilde{K}(E^\alpha, \kappa_\alpha(-\alpha^\vee) E^{-\alpha}) \\ &= +2 \tilde{K}(E^\alpha, E^{-\alpha}) \end{aligned}$$

$$\begin{aligned} \text{LHS} &= \tilde{K}(\frac{2}{|\alpha|^2} \alpha_i H^i, \frac{2}{|\alpha|^2} \alpha_j H^j) \\ &= \frac{4}{(|\alpha|^2)^2} \alpha_i \alpha_j \tilde{K}(H^i, H^j) \\ &= 4/|\alpha|^2 \end{aligned}$$

$$\therefore K(E^\alpha, E^{-\alpha}) = \frac{2}{|\alpha|^2} \quad \square$$

* The fundamental role of $\tilde{K}$ is to establish an isomorphism between the Cartan Subalgebra $\mathfrak{h}$ & its dual $\mathfrak{h}^*$

• $K(H^i, \cdot) : \mathfrak{h} \rightarrow \mathbb{R}$, fixed H^i

$\forall \gamma \in \mathfrak{h}^*, \exists H^\gamma \in \mathfrak{h} :$

$$\boxed{\gamma(H^i) := K(H^i, H^\gamma)}$$

* Let α be a root, then

$$H^\alpha := \alpha \cdot H = \alpha_i H^i$$

* W/ this we have a positive definite scalar product on $\mathfrak{h}^*$:

$$(X, Y) := K(K^X, K^Y)$$

$$\sim \text{For roots: } (\alpha, \alpha) = K(H^\alpha, H^\alpha) \\ = \alpha_i \alpha_i; K(H^i, H^i) = 1 \quad \checkmark$$

$$\therefore (\alpha, \alpha) = |\alpha|^2$$

Weights:

- Note, roots are associated to the roots of the adjoint rep of a group G . We would like to study the above type of system but for an arbitrary rep $\mathbb{R}$ of G . These eigenvalues are called the weights of the system.

Let $\mathbb{R}$ be spanned by its eigenstates $\{|\lambda\rangle\}$. Then the Cartan basis acts as:

$$H^i |\lambda\rangle = \lambda^i |\lambda\rangle$$

$$\lambda := (\lambda^1, \dots, \lambda^r) \quad (\text{Weights})$$

$$\lambda \in \mathfrak{g}^* \Leftrightarrow \lambda(H^i) = \lambda^i$$

→ The scalar product between weights is fixed by the Killing form

Since $[H^i, E^\alpha] = \alpha^i E^\alpha$, in this case the set $\{ |E^\alpha\rangle \equiv |\alpha\rangle \}$ were the basis of $(\mathfrak{g} \setminus \mathfrak{h})$. For a different rep, we use $\{ |\lambda\rangle \}$ s.t. $H^i |\lambda\rangle = \lambda^i |\lambda\rangle$.

$$\begin{aligned} * \text{Note: } H^i E^\alpha |\lambda\rangle &= [H^i, E^\alpha] |\lambda\rangle + E^\alpha H^i |\lambda\rangle \\ &= \alpha^i E^\alpha |\lambda\rangle + \lambda^i E^\alpha |\lambda\rangle \end{aligned}$$

$$\Rightarrow H^i (E^\alpha |\lambda\rangle) = (\alpha^i + \lambda^i) (E^\alpha |\lambda\rangle)$$

So, action of an element E^α on the basis of the rep raises the eigenvalue by a root!

If $E^\alpha \neq 0$, then naturally

$$E^\alpha |\lambda\rangle \propto |\lambda + \alpha\rangle$$

E^α : ladder operator

* For any $|\lambda\rangle$ in a finite dim rep, $\exists p, q \in \mathbb{Z}^+$

$$\begin{cases} (E^\alpha)^{p+1} |\lambda\rangle \propto E^\alpha |\lambda + p\alpha\rangle = 0 \\ (E^\alpha)^{q+1} |\lambda\rangle \propto E^\alpha |\lambda - q\alpha\rangle = 0 \end{cases}$$

- Note it is $(p+1)$ as there must be one "left" i.e. $E^{p+1}(E^\alpha |\lambda\rangle) \propto E^\alpha |\lambda + p\alpha\rangle$

$SL(2)$: Note that $\left\{ E^\alpha, E^{-\alpha}, \frac{\alpha \cdot H}{|\alpha|^2} \right\}$

For an $SL(2)$ subalgebra.

$$\begin{cases} E^\alpha =: J^+ \\ E^{-\alpha} =: J^- \\ \frac{\alpha \cdot H}{|\alpha|^2} =: J^3 \end{cases} \quad \begin{cases} [J^+, J^-] = 2J^3 \\ [J^3, J^\pm] = \pm J^\pm \end{cases}$$

* Let the dim of the subalgebra $SL(2)$ be $2j+1$

- $\dim(SL(2)) = 3$

but the above refers to the dim of the rep $\mathbb{R}$.

→ From the state $|\lambda\rangle$, the state w/ highest J^3 projection $m=j$ can be reached by a finite number, p , of app. of J^+

Applying J^- q -times leads to the same state w/ $m = -j$

$$H^i(J^+)^p |\lambda\rangle = (\lambda + (p-1)\alpha) J^+ |\lambda\rangle$$

$$J^3 |\lambda\rangle = (\alpha - H) / |\alpha|^2 |\lambda\rangle = \frac{(\alpha, \lambda)}{|\alpha|^2} |\lambda\rangle$$

$$\begin{aligned} J^3(J^+ |\lambda\rangle) &= [J^3, J^+] |\lambda\rangle + J^+ J^3 |\lambda\rangle \\ &= J^+ |\lambda\rangle + \frac{(\lambda, \alpha)}{|\alpha|^2} J^+ |\lambda\rangle \\ &= \left(\frac{(\lambda, \alpha)}{|\alpha|^2} + 1 \right) J^+ |\lambda\rangle \end{aligned}$$

$$\therefore J^3(J^+)^p |\lambda\rangle = \left(\frac{(\lambda, \alpha)}{|\alpha|^2} + p \right) J^+ |\lambda\rangle$$

~> Note, that the dim is $2j+1$

Since

$$\begin{array}{c} \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \\ -j \quad \leftarrow \quad 0 \quad \xrightarrow{p} \quad j \end{array}$$

So, if we start w/ $|\lambda\rangle$ then move up to j via p .

$$\text{then } J^3(|\tilde{\lambda}\rangle) = j |\tilde{\lambda}\rangle$$

$$\text{So } \boxed{j = \frac{(\alpha, \lambda)}{|\alpha|^2} + p} \Leftrightarrow \boxed{-j = \frac{(\alpha, \lambda)}{|\alpha|^2} - q}$$

$$\text{or, } \boxed{2 \frac{(\alpha, \lambda)}{|\alpha|^2} = -(p - q)} \quad \text{⊗}$$

=> Any weight λ in a finite-dim rep is such that $\boxed{(\alpha, \lambda) / |\alpha|^2 \in \mathbb{Z}}$

Simple Roots in the Cartan Matrix:

$$\begin{aligned}\# \text{ roots} &= \dim(\mathfrak{g}) - \text{rk}(\mathfrak{g}) \gg \text{rk}(\mathfrak{g}) \\ &= \dim(\mathfrak{g}) - \dim(\mathfrak{h})\end{aligned}$$

• So roots are in general linearly dep.

→ Thanks to the $\mathbb{R}$ inner product

$$(\gamma, \beta) = \tilde{K}(H^\gamma, H^\beta) \text{ on } \mathfrak{h}^*$$

So we can interpret the roots α geometrically. I.E. as a "root vector"

→ Introduce a hyperplane in the root space Δ which is mutually orthogonal to all roots. This hyper plane will split Δ into:

$$\Delta = \Delta_+ \oplus \Delta_-$$

$$\left\{ \begin{array}{l} \Delta_+ = \text{Positive roots} \\ \Delta_- = \text{Negative roots} \end{array} \right\} \quad \boxed{\Delta_- = -\Delta_+}$$

* Such a split is NOT unique. It is completely basis dependant.

$$\text{Define: } \boxed{\mathfrak{g}_\pm := \text{span} \{ E^{\pm \alpha}, \alpha > 0 \}}$$

⇒ Subalg. spanned by the step ops.

$$\boxed{\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{g}_+ \oplus \mathfrak{g}_-}$$

↑

~~→~~ Cartan (triangular) decomposition

$\left\{ \begin{array}{l} E^\alpha: \text{Raising op.} \\ E^{-\alpha}: \text{Lowering op.} \end{array} \right\}$

Def: A simple root α_i is defined to be a root that cannot be written as a sum of two positive roots. i.e. it is linearly independent in the space of roots.

Geometrically: A simple root are those positive roots that are closest to the hyperplane used to sep. Δ^+ & Δ^-

$\rightarrow$ Since $\text{ad}_{H_i}(E^\alpha) = \alpha_i E^\alpha$ must be unique for all α , these must be the simple roots.

$$\Rightarrow r = \text{rk}(\mathfrak{g}) = \dim(\mathfrak{h}) = \dim(\mathfrak{h}^*)$$

r : Simple roots

$$\Delta_s := \{\alpha_1, \dots, \alpha_r\}$$

where α_i is a labeling of the simple root α_i at $i \in \text{index}$.

* This is the most natural basis to use in Δ .

From the above definition

- $\alpha_{ci} - \alpha_{ci} \notin \Delta$

- any positive root is a sum of positive root

($\Rightarrow$) Δ as a basis is not orthonormal

Cartan-Matrix: $A_{(ij)} \equiv 2(\alpha_{ci}, \alpha_{cj}) / |\alpha_{cj}|^2$

Note, this just $\frac{2(\alpha_{ci}, \alpha_{cj})}{|\alpha_{cj}|^2} = -(p-q)$
from earlier.

which summarizes the structure of a
semi-simple complex LA.

Note $A_{ii} = 2(\alpha_{ci}, \alpha_{ci}) / |\alpha_{ci}|^2 = 2$

and, in general $A_{(ij)} \neq A_{(ji)}$

- Cauchy-Schwarz:

$$|\langle A, B \rangle|^2 \leq \|A\|^2 \|B\|^2$$

$$\Rightarrow |(\alpha_{ci}, \alpha_{cj})|^2 \leq \|\alpha_{ci}\|^2 \|\alpha_{cj}\|^2$$

Since $A_{ij} = 2(\alpha_{ci}, \alpha_{cj}) / |\alpha_{cj}|^2$

$$\Rightarrow (\alpha_{ci}, \alpha_{cj})(\alpha_{cj}, \alpha_{ci}) \leq |\alpha_{ci}|^2 |\alpha_{cj}|^2$$

$$\Rightarrow A_{ij} A_{ji} \left(\frac{1}{4}\right) \leq 1$$

$$A_{ij} A_{ji} \leq 4, \quad i \neq j$$

$$A_{ii} A_{ii} = 4, \quad i = j$$

Note $A_{ij} = m$, $m \in \mathbb{Z}$

$$\bullet A_{ij} A_{ji} = m \cdot n < 4$$

$$m, n < 4 \quad \text{but } m, n \in \mathbb{Z}^+$$

$$\bullet \text{So } m = 1, n = 1, 2, 3$$

↳ likewise for $n = 1, \dots$

$$\bullet \boxed{A_{ii} = 0, -1, -2, -3}$$

* For the set of roots two lengths are possible. "long" or "short"

$$\boxed{|\alpha_L|^2 / |\alpha_S|^2 = 2 \text{ or } 3}$$

Def: If all the roots in Δ_S have the same length, i.e.

$$A_{ij} = 2(\alpha_i, \alpha_j) / |\alpha_j|^2 = A_{ji}$$

↳ the algebra is called Simply Laced.

Def: The coroot associated to a root α_i is defined as

$$\boxed{\alpha_i^V = 2\alpha_i / |\alpha_i|^2}$$

The scalar product between a root α_i & coroot is always an int $\mathbb{Z}$

$$\boxed{A_{ij} = (\alpha_j, \alpha_i^V)}$$

The highest Root Θ is the unique root for which $\sum_i m_i$ is maximized in the expression $\sum_i m_i \alpha_i$.

- All elements of Δ can be reached by repeated subtraction of simple roots from Θ

$$\Theta = \sum_{i=1}^r a_i \alpha_i = \sum_{i=1}^r a_i^V \alpha_i^V, \quad a_i, a_i^V \in \mathbb{N}$$

a_i are called marks or Kac-labels.
 a_i^V comarks or dual-Kac-labels

Note RPS: $\sum_{i=1}^r a_i^V 2 \alpha_i / |\alpha_i|^2$
 $\Rightarrow a_i = 2 a_i^V / |\alpha_i|^2$

The dual Coxeter number is defined as $g = \sum_{i=1}^r a_i^V + 1$

which normalizes $K(X, Y)$

$$K(X, Y) = \frac{1}{2g} \text{Tr}(\text{ad}(X) \circ \text{ad}(Y))$$

The Chevalley basis:

* We can obtain the simple roots from the Cartan Matrix.

Chevalley basis: To each simple root corresponds to three generators:

$$e^i = E^{\alpha_i}, \quad f^i = E^{-\alpha_i}, \quad h^i = \frac{2\alpha_i \cdot H}{|\alpha_i|^2}$$

So the commutation relations:

$$\begin{aligned} [h^i, h^j] &= 0 \\ [h^i, e^j] &= 2(\alpha_i)_k (H^k, e^j) / |\alpha_i|^2 \\ &= 2(\alpha_i)_k \alpha_{ij}^k e^j / |\alpha_i|^2 \end{aligned}$$

$$[h^i, e^j] = A_{ji} e^j$$

$$[h^i, f^j] = -A_{ji} f^j$$

$$[e^i, f^j] = \delta_{ij} h^j$$

* All the structure constants in the Chevalley basis are integers.

All remaining step ops are obtained by repeated commutation relations of the basic generators.

Serre Relations:

$$\begin{aligned} [ad(e^i)]^{1-A_{ji}} e^j &= 0 \\ [ad(f^i)]^{1-A_{ji}} f^j &= 0 \end{aligned}$$

* These are main results that we will use.

Note:

$$K(h^i, h^j) = (\alpha_i^V, \alpha_j^V)$$

Dynkin diagrams:

To each Cartan matrix, a diagram made of vertices $\hat{e}_i$ conn. lines

(i) Each vertex in the diag. represents a simple root. (# vertices = $\text{rk}(\mathfrak{g})$)

(ii) Long roots $\circ$

Short roots $\bullet$

(iii) Vertices i, j are joined by $A_{ij}A_{ji}$ lines

— Ortho. roots corresponds to disjoint vertices

— Vert. connected by one line

Correspond to simple roots

— " spanning an angle of $\frac{2\pi}{3} = 120^\circ$

— " "

two lines

" $\frac{3\pi}{4} = 135^\circ$

— " "

three lines

" $\frac{5\pi}{6} = 150^\circ$

Ex: For $\text{rk}(A) = 2 \Rightarrow \dim(V) = 2$

The Cartan Matrix is 2×2

$$A_{ij} A_{ji} \leq 4, \quad A_{ii} = 2(\alpha_i, \alpha_i) / |\alpha_i|^2$$

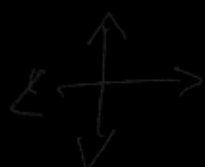
$$A_{11} = 2, \quad A_{22} = 2$$

These are the possible values of A :

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$0 \quad 0$$

$$A_1 \oplus A_2$$



$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

$$0 \quad 0$$

$$A_2$$



$$\begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}$$

$$0 \quad \bullet$$

$$B_2$$



$$\begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$$

$$0 \quad \bullet \bullet$$

$$G_2$$



Cartan Classification for finite dim.

Simple LA:

$$A_r (r \geq 1) \quad B_n (n \geq 3) \quad C_r (r \geq 2), \quad D_r (r \geq 4)$$

$$E_6, E_7, E_8, G_2, F_4 \quad (\text{Exceptional LA's})$$

• The subscript denotes the rank of the group.

$$A_r \cong \mathfrak{sl}(r+1)$$

$$B_r \cong \mathfrak{so}(2r+1)$$

$$C_r \cong \mathfrak{sp}(r)$$

$$D_r \cong \mathfrak{so}(2r)$$

$$A_1 \cong B_1 \cong D_1 \cong C_1$$

$$B_2 \cong C_2$$

$$D_2 \cong A_1 \oplus A_1$$

$$D_3 \cong A_3$$

$$D_r = 2A(r-1)$$

$$D_3 = A_3$$

A_r, D_r, E_6, E_7, E_8 : Simply laced

B_r, C_r, F_4 , (long) = $\sqrt{2}$ (short)

A_r : $\begin{array}{c} \circ - \circ - \circ \dots \circ - \circ \\ (1) (2) \dots (r+1) (r) \end{array}$

B_r $\begin{array}{c} \circ - \circ \dots \circ \rightarrow \bullet \\ (1) (2) \dots (r-1) (r) \end{array}$

C_r $\begin{array}{c} \bullet - \bullet \dots \bullet \leftarrow \bullet \\ 1 \quad 2 \quad \quad \quad r-1 \quad r \end{array}$

D_r $\begin{array}{c} \circ - \circ - \circ \dots \circ \begin{array}{l} \nearrow \circ^r \\ \searrow \circ^{r-1} \end{array} \\ 1 \quad 2 \quad 3 \quad \quad \quad r-2 \end{array}$

E_6 : $\begin{array}{c} \circ^6 \\ | \\ \circ - \circ - \circ - \circ - \circ \\ 1 \quad 2 \quad 3 \quad 4 \quad 5 \end{array}$

E_7 : $\begin{array}{c} \circ^7 \\ | \\ \circ - \circ - \circ - \circ - \circ - \circ \\ 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \end{array}$

E_8 : $\begin{array}{c} \circ^8 \\ | \\ \circ - \circ - \circ - \circ - \circ - \circ - \circ \\ 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \end{array}$

F_4 : $\begin{array}{c} \circ - \circ \rightarrow \bullet \leftarrow \bullet \\ 1 \quad 2 \quad 3 \end{array}$

G_2 : $\begin{array}{c} \circ \rightarrow \bullet \leftarrow \bullet \\ 1 \quad 2 \end{array}$

Fundamental Weights & Dynkin labels =

* Weights can be expanded in the basis of simple roots as we saw
 for example: $H^i(E^\lambda/\lambda) = (\lambda^i + \alpha^i) E^{\alpha^i}$

→ For irreducible finite-dim rep, the coeffs are not integers.

We choose the dual of the simple coroot basis (α_{ci}^V) .

Recall, on the space of weights we have an isomorphism from $\mathfrak{h} \rightarrow \mathfrak{h}^*$
 so the dual basis $\{\omega_i\}$ of the coroot is defined by

$$(\omega_{ci}, \alpha_{cj}^V) = \delta_{ij} = \omega_{ci}(\alpha_{cj}^V) \\
= (\omega_{ci}, 2\alpha_{cj}/|\alpha_{cj}|^2) = \frac{2}{|\alpha_{cj}|^2} (\omega_{ci}, \alpha_{cj})$$

Fundamental weights: $\{\omega_i, i=1, \dots, r\}$

This set is the Dynkin Basis

Because we have a basis of the weights: $\{\omega_{ci}\} \subset \Delta$

$$\lambda = \sum_k \lambda_k \omega_{ck} \Leftrightarrow (\lambda, \alpha_{ci}^V) = \lambda_i$$

λ^i : Dynkin labels



* Dynkin labels of weights in a finite dim irrep. are always integers.
 $\hookrightarrow$ A weight w/ integer expansion in $\{\omega_i\}$ is called integral.

$$\lambda = (\lambda_1, \dots, \lambda_r)$$

$\rightarrow$ It should be understood, from hence forth, as Dynkin labels

$$\lambda = [\lambda_1, \dots, \lambda_r]$$

* Notation I will use!

Now, to relate this to the set of roots we note, from above,

$$\frac{2}{(\alpha_{ci})^2} (w_{ci}, \alpha_{ci}) = \delta_{ci}$$

$$\Rightarrow A_{ij} (w_{ci}, \alpha_{ci}) = (\alpha_{ci}, \alpha_{ci}) \delta_{ij}$$

$$\Rightarrow (A_{ij} w_{ci}, \alpha_{ci}) = (\alpha_{ci}, \alpha_{ci})$$

$$\Rightarrow (\alpha_{ci} - \sum_{i=1}^r A_{ij} w_{ci}, 0) = 0$$

Since, $(\cdot, \cdot)$ is non-degenerate we find that

$$* \boxed{\alpha_{ci} = \sum_{i=1}^r A_{ij} w_{ci}} *$$

$\Rightarrow$ The i^{th} -row of A is the set

of Dynkin labels for a simple root α_i

$$(A_i)^j = \lambda^j$$

ex: $A_2: \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$

$$\alpha_1 = 2\omega_1 - 1\omega_2$$

$$\alpha_2 = -1\omega_1 + 2\omega_2$$

* Dynkin labels are the eigenvalues of the Chevalley gen's of the Cartan Subalg:

$$\begin{aligned} h^i |\lambda\rangle &= \lambda(h^i) |\lambda\rangle = (\lambda, \alpha_{(i)}^V) |\lambda\rangle \\ h^i |\lambda\rangle &= \lambda_i |\lambda\rangle \end{aligned}$$

- λ_i : eigenvalue of h^i (Dynkin label)
- λ_i : eigenvalue of H^i

Suppose $\lambda^1 = \lambda^2 = \dots = \lambda^r = 1$ for ρ
 $\rho = \sum_i \omega_i = (1, 1, \dots, 1) = [1, 1, \dots, 1]$
 (Weyl vector)

→ Also defined as

$$\rho = \frac{1}{2} \sum_{\alpha \in \Delta_+} \alpha$$

The scalar product of weights can be expressed as

$$(w_i, w_j) = F_{ij} = F(w_i, w_j)$$

F_{ij} is the matrix relating $\{w_i\}$

the simple coroot basis $\{\alpha_i^\vee\}$

$$\boxed{\omega_i = \sum_{j=1}^r F_{ij} \alpha_j^\vee}$$

$$F_{ij} = (A^{-1})_{ij} \frac{|\alpha_j|^2}{2}$$

* Given two weights $\lambda = \sum \lambda_i \omega_i, \mu = \sum \mu_i \omega_i$

$$(\lambda, \mu) = \sum_{ij} \lambda_i \mu_j F_{ij}$$

The Weyl Group:

* The roots in a simple LA. have a high degree of Cartan dual symmetry.

($\rightarrow$) $\text{Aut}(\Delta)$: Automorphism group of Δ
i.e. $\text{Aut}(\Delta) = \{\phi: \Delta \rightarrow \Delta, \text{ s.t. } \exists \phi^{-1}\}$

* Note, $\exists$ a $\mathbb{Z}_2$ symmetry since, if $\alpha \in \Delta$, then so too does $-\alpha$
 $\alpha \mapsto -\alpha, \alpha \in \Delta$

$\rightarrow \text{Aut}(\Delta) \subset S_{\dim(\mathfrak{g}) - r}$

Where S_n is the symmetric group w/ $\dim(\mathfrak{g}) - r$ elements.

* Reflection wrt. the hyperplane (through 0) in the root space

perp. to a fixed root α :

Denote this action S_α

$$S_\alpha(\beta) := \beta - 2 \frac{(\beta, \alpha)}{(\alpha, \alpha)} \alpha$$

ex: 

- We are subtracting twice $\hat{\alpha}$ (proj.) which is simply a permutation of roots.

$W \equiv$ Weyl Group

$$W := \{ S_{\alpha_1}, \dots, S_{\alpha_r} \}$$

This group has the binary op.

$$S_\alpha \circ S_{\alpha'} = S_{\alpha'} \circ S_\alpha$$

So W is an abelian subgroup of $S_{\dim(\mathfrak{g}) - r}$.

$\rightarrow$ Note $S_\alpha \circ S_\beta$ yields to reflections and rotations, so a generic element of W will not be of the form

$$S_\alpha \beta = \beta - 2 \frac{(\alpha, \beta)}{(\alpha, \alpha)} \alpha = \beta - (\alpha, \beta) \alpha^\vee$$

$\Rightarrow$ By linearity, the Weyl group action

extends to weights:

$$\begin{aligned} S_{\alpha}(\sum_{i,j} A_{ij} \lambda^{ij}) &= \sum_{i,j} A_{ij} S_{\alpha}(\lambda^{ij}) = 2(\alpha, \sum_{i,j} A_{ij} \lambda^{ij}) \\ &= \sum_{i,j} A_{ij} 2(\alpha, \lambda^{ij}) = 2(\alpha, \sum_{i,j} A_{ij} \lambda^{ij}) \\ &= \sum_{i,j} A_{ij} S_{\alpha}(\lambda^{ij}) \end{aligned}$$

$$\Rightarrow \boxed{S_{\alpha} \lambda = \lambda - (\alpha^{\vee}, \lambda) \alpha}$$

* W is an isometry group on the space of weights

$$\boxed{(S_{\alpha} \lambda, S_{\alpha} \omega) = (\lambda, \omega)}$$

Proof:

$$\begin{aligned} &(\lambda - (\alpha^{\vee}, \lambda) \alpha, \omega - (\alpha^{\vee}, \omega) \alpha) \\ &= (\lambda, \omega) - (\alpha^{\vee}, \omega)(\lambda, \alpha) - (\alpha^{\vee}, \lambda)(\alpha, \omega) \\ &\quad + (\alpha^{\vee}, \lambda)(\alpha^{\vee}, \omega)(\alpha, \alpha) \\ &= (\lambda, \omega) - \frac{2}{|\alpha|^2}(\alpha, \omega)(\lambda, \alpha) - \frac{2}{|\alpha|^2}(\alpha, \lambda)(\alpha, \omega) \\ &\quad + (\alpha, \lambda)(\alpha, \omega)(\alpha^{\vee}, \alpha^{\vee}) \end{aligned}$$

$$\text{But } (\alpha^{\vee}, \alpha^{\vee}) = \frac{4}{|\alpha|^4}(\alpha, \alpha) = \frac{4}{|\alpha|^2}$$

$$|\alpha|^2 = (\alpha, \alpha)$$

So

$$\begin{aligned} &= (\lambda, \omega) - \frac{2}{|\alpha|^2}(\alpha, \omega)(\lambda, \alpha) - \frac{2}{|\alpha|^2}(\alpha, \lambda)(\alpha, \omega) \\ &\quad + (\alpha, \lambda)(\alpha, \omega) \frac{4}{|\alpha|^2} \\ &= (\lambda, \omega) \checkmark \end{aligned}$$

Let α_i be a simple root. Then

$$\boxed{S_{\alpha_i} \equiv S_i} \quad i=1, \dots, r$$

(Simple/fundamental
Weyl reflection)

Then the fundamental Weyl reflects
generate the Weyl group W .

* Every $w \in W$ can be written as a
word (Weyl word)

$$\boxed{w = S_{i_1} S_{i_2} \dots S_{i_k}}$$

in "letters" S_i

$\rightarrow$ This is not unique.

def: The length of $w = (\# \text{ of } (S_i)'s)$
 $l(w) = k$

The signature of w is defined as

$$\boxed{\epsilon(w) := (-1)^{l(w)}}$$

$$\boxed{S_i^2 = \mathbb{1}, S_i S_j = S_j S_i \text{ if } A_{ij} = 0}$$

Or, more generally,

$$\boxed{(S_i S_j)^{m_{ij}} = \mathbb{1}}$$

$$\boxed{m_{ij} := \begin{cases} 2, & i=j \\ \frac{\pi}{\pi - \theta_{ij}}, & i \neq j \end{cases}}$$

$\theta_{ij} \equiv$ angle between roots α_i, α_j

$$\cos(\theta_{ij}) = \frac{(\alpha_i, \alpha_j)}{|\alpha_i| |\alpha_j|}$$

$$\cos(\theta_{ij}) = \frac{1}{2|\alpha_i|^2} A_{ij}$$

$$\Rightarrow \boxed{\theta_{ij} = \cos^{-1}\left(\frac{A_{ij}}{2|\alpha_i|^2}\right)}$$

"□"

This is the defining relation of the Weyl group (ref for w_i)

Def: Any group having (such?) such a rep. is called a Coxeter group.

Note, for simple roots $\alpha_{(i)}, \alpha_{(j)}$

$$\boxed{S_i \alpha_{(j)} = \alpha_{(j)} - A_{ji} \alpha_{(i)}} \leftarrow \begin{cases} \text{NO} \\ \text{Sum over } i \end{cases}$$

$$\Delta = \{w_{i_1} \alpha_{i_1}, \dots, w_{i_n} \alpha_{i_n} \mid w_{i_{(a)}} \in W\}$$

$\Rightarrow$ The set $\{w' \alpha_i\}$ can, for fixed w' could serve as a basis of Δ_s .

Weyl group induces a natural splitting of the r -dim weight vector space into a fan of open cones C_w

These cones, whose # is equal to $|W|$ are defined by

$$\{ \lambda \in \Lambda \mid (\lambda, \alpha_{ci}) \geq 0, i=1, \dots, r, w \}$$

Weyl-Chambers

$$\text{If } (\lambda, \alpha_{ci}) = \cos(\tilde{\theta}_{ci}) |\lambda|^2 |\alpha_{ci}|^2 \geq 0 \Rightarrow \cos(\tilde{\theta}) \geq 0$$

$$\Leftrightarrow \theta_{ci} \in [-\pi/2, \pi/2]$$

wrt. each α_{ci}

* They intersect only at the reflecting hyperplanes, their boundaries $(\lambda, \alpha_{ci}) = 0$ i.e. orthogonal.

One of the chambers, depending on α_{ci} is distinguished: The unique chamber whose pts. have only positive Dynkin labels $\lambda_i \in \mathbb{Z}$.

→ This is the fundamental or dominant Weyl chamber C_0

→ Weight in this chamber is called dominant. θ : Highest root is an example.

→ Acting with W on $C_0 + \partial C_0$ one obtains the entire root space.

* Normalization convention:

sq root of the long root is set to 2
 $|\theta|^2 = 2$

ex: Rank 2 root systems

(i) $A_2: \mathfrak{sl}(3)$ $\circ \text{---} \circ$ $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$

$\alpha_1 = (2, -1)$, $\alpha_2 = (-1, 2)$

The quadratic form Q :

$F_{ij} = (A^{-1})_{ij} \frac{|\alpha_i|^2}{2} = \frac{1}{2} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

So $W = \{1, s_1, s_2\}$

$(\alpha_1, \alpha_2) \Rightarrow \cos(\theta) \frac{|\alpha_1|^2 + |\alpha_2|^2}{2} = -4/5$

$\theta = \frac{2\pi}{3}$

$(s_1 s_2)^3 = 1$

$(s_1 s_2 s_1 s_2 s_1 s_2) = 1$

$(s_2 s_1 s_2)^{-1} = s_1 s_2 s_1$

$(s_2 s_1 s_2)(s_2 s_1 s_2)^{-1} = s_2 s_1 s_2 s_1 s_2 s_1 = 1$

$\Leftrightarrow s_2 s_1 s_2 = s_1 s_2 s_1$

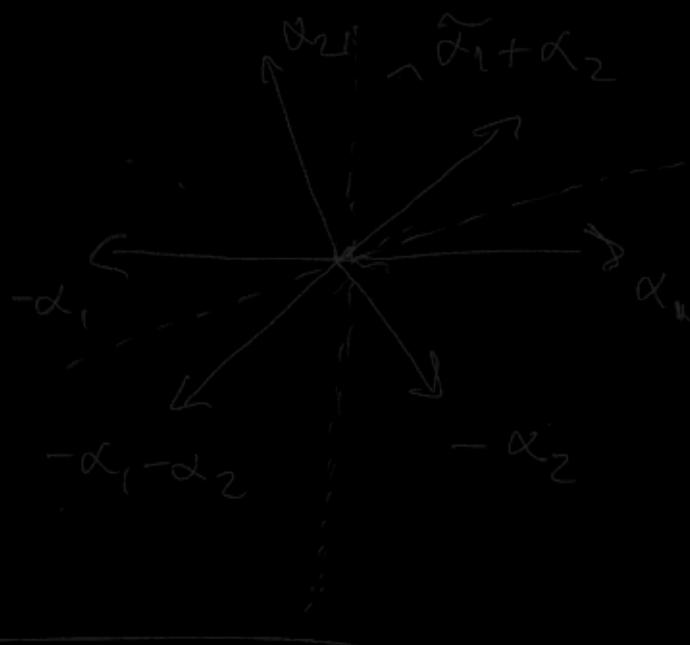
So no words w/ more than 3 elements exist.

$W = \{1, s_1, s_2, s_1 s_2, s_2 s_1, s_1 s_2 s_1\}$

$\Delta = \{\alpha_1, \alpha_2, \alpha_1 + \alpha_2, -\alpha_1 - \alpha_2, -\alpha_1, -\alpha_2\}$

→ Highest root $\alpha_1 + \alpha_2$

* Note



$[cc] C_2$ ω_1, ω_2 $\neq 0$, $\alpha_1 = 2\omega_1 - \omega_2$
 $E = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ $A = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}$ $\alpha_2 = -2\omega_1 + 2\omega_2$

W. generated by $\underline{1}, s_1, s_2$

$\theta_{12} = 3\pi/4$: $(\alpha_1, \alpha_2) = \frac{1}{2} A_{12} |\alpha_2|^2$
 $= 2(-\frac{1}{2}) |\alpha_2|^2 = \cos(\theta) |\alpha_1|^2 |\alpha_2|^2$
 $\cos(\theta) = -1/|\alpha_1|^2 = -1/5$

* $(\alpha_1, \alpha_2) = \cos(\theta) |\alpha_{\alpha_1}|^2 |\alpha_{\alpha_2}|^2$
 $\frac{1}{2} A_{12} |\alpha_2|^2 = \cos(\theta) |\alpha_{\alpha_1}|^2 |\alpha_{\alpha_2}|^2$
 $= 1 = \cos(\theta) |\alpha_{\alpha_1}|^2$
 $-1/\sqrt{2} = \cos(\theta)$

$\frac{\sqrt{2}}{2}$ $\theta = \pi - \pi/4 = \frac{3\pi}{4}$

→ Normalize $|\alpha_2|^2 = 2$

$$(S_1 S_2)^2 = I$$

$$S_1 S_2 S_1 S_2 = S_2 S_1 S_2 S_1$$

$$W = \{I, s_1, s_2, s_1 s_2, s_2 s_1, s_1 s_2 s_1, s_2 s_1 s_2, s_1 s_2 s_1 s_2\}$$

Highest root: $\Theta = 2\alpha_1 + \alpha_2$
 $= 2(2\omega_1 - \omega_2) + (-2\omega_1 + 2\omega_2)$
 $= 4\omega_1 - 2\omega_2 - 2\omega_1 + 2\omega_2$
 $= 2\omega_1$

Visualizing the root system of higher rank

Simple LA's:

* The proj. of choice for root system is the one onto the Coxeter plane.

The coxeter element of a L.G. is the product of all simple weyl ref.

$$\rightarrow S_{i_1} \circ \dots \circ S_{i_r} = w^\vee$$

Note, changing the order of the reflections leads to a conj. Coxeter element.

The coxeter # h is the # of roots divided by the rank of the alg.

$$h = \dim(\mathfrak{a})/r$$

Highest Weight Reps:

$$A_1: \mathfrak{sl}(2) \cong \mathfrak{su}(2)$$

We look at this subalgebra to understand reps. So each simple root has an associated $\{E^\alpha, E^{-\alpha}, H^\alpha\} \mathfrak{sl}(2, \mathbb{C})$ alg.
 $\Rightarrow$ Any rep has a basis on which the whole Cartan subalgebra $\mathfrak{h}$ acts diagonally.

* Any finite dim irrep has a unique highest weight state $|\lambda\rangle$

which is completely specified by its

$$= \text{Dynkin labels } \boxed{\lambda(H^i) = \lambda_i}$$

$\sum_i \lambda_i$ is maximal ~~among~~ amongst all other weights when expanded in terms of $\lambda = \sum_i \lambda_i \omega^i \Rightarrow (A_i)^i = \lambda_i$

Thus, if $\alpha > 0$, $\lambda + \alpha$ cannot be a weight in the rep. so that

$$\boxed{E^\alpha |\lambda\rangle = 0}, \quad \forall \alpha > 0$$

In this case $p=0$, it is also therefore dominant.

* An irreducible finite-dim. rep space of a semi-simple Lt. is characterized by the fact that it has a highest weight of ~~the~~ multiplicity one and that this weight is dominant integral

$$\hookrightarrow \lambda_i \in \mathbb{Z}$$

* Representations are often labelled by their highest weight.

Highest weight of adj. : $\underline{\theta} =$

$L_\lambda := (\text{Unique, }^{finite} \text{irrep of highest weight } \lambda)$

Weights & Multiplicities:

Starting from $|\lambda\rangle$ all other elements in L_λ can be obtained by the action $E^{-\beta} E^{-\gamma} \dots E^{-\alpha} |\lambda\rangle$, for $\beta, \gamma, \dots, \alpha \in \Delta_+$

The set of eigenvalues of all states in L_λ is called the Weight System

$$\boxed{\Omega_\lambda}$$

Any $\lambda \in \Delta_+$ s.t. $\lambda - \lambda \in \Delta_+$

$\Rightarrow$ All the weights of a given rep. lie in exactly one congruence class.

Let

$$\lambda' = \lambda - \sum_i n_i \alpha_i, \quad n_i \in \mathbb{Z}^+$$

$$\sum n_i \equiv (\text{depth/level}).$$

If $\lambda - \mu = \beta, \quad \beta = \sum_i n_i \alpha_i$
then $\boxed{\mu \leq \lambda}$

This gives a partial ordering of Ω_λ which again $\sim \{\sum n_i \alpha_i\}$

$\hookrightarrow$ Any finite irrep $L_\lambda \exists$ weight of depth 0 ($\sum n_i = 0 \Rightarrow n_i = 0 \forall i$)
namely, the highest weight λ .

To find all $\lambda' \in \Omega_\lambda$, Recall

$$2 \frac{(\alpha, \lambda)}{(\alpha, \alpha)} = 1 - p$$

$$\hookrightarrow (w\alpha, \alpha_i^\vee) = \delta_{i, w^{-1}(i)}$$

$$(\alpha_{(i)}^\vee, \lambda') = -(p_{(i)} - q_{(i)})$$

$$\lambda' = \sum_k \lambda_k' w_k$$

$$\boxed{\sum_k \lambda_k' (\alpha_{(i)}^\vee, w_k \alpha) = \lambda_{(i)}' = -(p_{(i)} - q_{(i)})}$$

So that

$$\boxed{\lambda' \equiv \lambda - \sum_i n_i \alpha_i}$$

$\hookrightarrow$ To find λ' we proceed level-by-level
* When do we terminate? Until we

arrive at a depth where all the weights have negative Dynkin labels of the same rep.

ex: A_2 Adj + $\lambda = (1, 1)$

$0-0 \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \alpha_1 = (2, -1), \alpha_2 = (-1, 2)$

$\begin{pmatrix} 1 & 1 \\ \downarrow -\alpha_1 & \rightarrow \alpha_2 \end{pmatrix}$

$\dim(K_1) = 10?$

$\begin{pmatrix} -1 & 2 \\ \downarrow \alpha_2 & \swarrow \alpha_1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ \swarrow \alpha_1 & \downarrow \alpha_2 \end{pmatrix}$

$\square: \frac{2 \cdot 3 - 1}{3} = 2$

$\hookrightarrow L(2)$ adjoint

$\dim A_2 = 16$

$\begin{pmatrix} 0 & 0 \\ \swarrow \alpha_1 & \searrow \alpha_2 \end{pmatrix}$
 $\begin{pmatrix} -2 & 1 \\ \downarrow & \swarrow \end{pmatrix} \begin{pmatrix} 1 & -2 \\ \swarrow & \downarrow \end{pmatrix}$
 $\begin{pmatrix} -1 & -1 \end{pmatrix}$

Note "l-o-p-o"

$\left. \begin{array}{l} A_r: SL(r+1) \\ B_r: SO(2r+1) \\ C_r: Sp(r) \\ D_r: SO(2r) \end{array} \right\} \begin{array}{l} \Rightarrow \\ \Leftarrow \\ \circ \end{array}$

II-d Sugra: Massless reps
 have a little group of $SO(4)$

$\Rightarrow$ B4: $\overline{L}_{int} = W_m, \overline{L}_{int} = m W_{\pm}$

W_m : fundamental spinor. $n=4$

$$\therefore \{ \chi_{\mu\alpha} \}, (A\overline{MNP}), (g\overline{MN})\}$$

$$[1001]_9, [0100]_9, [2000]_9$$

Type IIA & Type IIB 10-d Supers:

Type IIA:

$$([1000]_8 \oplus [0010]_8) \otimes ([1000]_8 \oplus [0001]_8)$$

$\rightarrow$ Has $N = (2, 1)$ 8 vgrs, non-chiral

Type IIB: Chiral $N = (2, 0)$

$$([1000]_8 \oplus [0001]_8) \otimes ([1000]_8 \oplus [0001]_8)$$

The multiplicity of a (non-highest) weight must be larger than one for certain reps.

$\rightarrow$ Require: $\frac{1}{2}(\dim \mathfrak{g}) - r = |\Delta + 1|$ labels.

• $\frac{1}{2}(\dim \mathfrak{g}) - 3r$ quantum #s in addition to the Dynkin labels.

Frobenius recursion formula:

* Multiplicities of λ' in the rep λ , in terms of all the weights above it:

$$(|\lambda + \rho|^2 - |\lambda' + \rho|^2) \text{mult}_\lambda(\lambda') = 2 \sum_{\alpha \neq 0} \sum_{k=1}^{\infty} (\lambda' + k\alpha, \alpha)$$

$$\text{mult}_\lambda(\lambda' + k\alpha)$$

→ ex: Mult. of $(0,0)$ in the adjoint of A_2

$$k=1: (\lambda' + k\alpha, \alpha) = 2$$

$$\lambda = \theta = \rho = \alpha_1 + \alpha_2$$

$$(8-2) \text{mult}_\theta(0,0) = 2(2+2+2)$$

$$\Rightarrow \text{mult}_\theta(0,0) = 2$$

* Ad_g° : Will have mult 1 for all roots except $\lambda=0$, w/ mult. = r_+

* Finite-dim irrep L_λ is always unitary
 $(H^i)^\dagger = H^i, (E^\alpha)^\dagger = (E^{-\alpha})$
 $\Rightarrow \langle \lambda | \lambda \rangle \geq 0$ for $|\lambda\rangle \in L_\lambda$

Recall: $(\omega_i, \alpha_j^\vee) = \delta_{ij}$ $\{\omega_i\} = \text{fund. weights}$
 $\lambda = \sum_i \lambda_i \omega_i$ λ_i : Dynkin labels $|\alpha_i^\vee| = 2\alpha_i/|\alpha|^2$

$$\Leftrightarrow \alpha_i = \sum_j A_{ij} \omega_j$$

Def: The W-orbit of a weight λ is $\{w\lambda \mid w \in W\}$
 • The W-orbit of every weight has exactly one pt in C_0

Conjugate Reps:

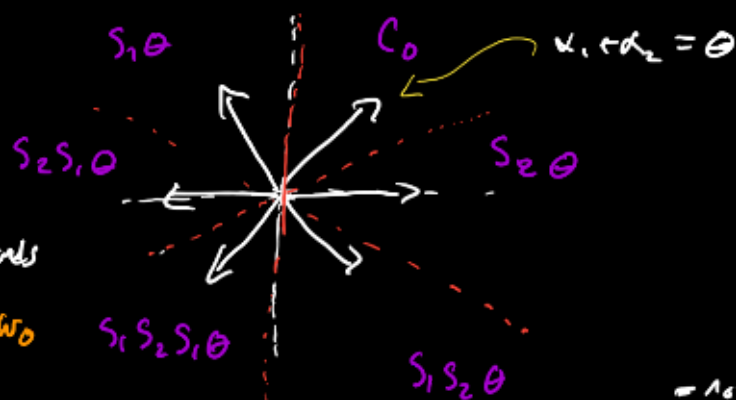
• Each irrep has a lowest weight state $(-\theta)$. This state lies

in the W -orbit of the weight λ , in the Weyl Chamber opposite C_0 .

ex: A_2 

* This Chamber corresponds to the longest element w_0 in W

$\Rightarrow$ Lowest weight state: $w_0 \lambda$



$\rightarrow$ not $\approx$ since not DUAL

$\rho^*: G \rightarrow \text{Aut}(V)$

Def: The conjugate rep: λ^* has for its highest weight the negative of the lowest weight state λ of L_λ

$$\lambda^* = -(w_0 \lambda)$$

• The Weyl vector ρ is the highest weight of the self conj. rep:

$$\rho = -(w_0 \rho)$$

More generally, Ω_λ^* are all the negatives of those in Ω_λ .

$\leadsto$ Conj. related to reflection of the Dynkin Diagram

A_r : 

* conj. amounts to reversing the order of the Dynkin labels.

Def: A rep. is self-conj. iff its weight system is invariant under a change of sign.

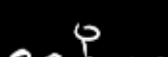
• For $\alpha \in \Delta_+$ $\exists \tilde{\alpha} \in \Delta_-$ s.t. $\alpha = -\tilde{\alpha}$ (obviously)

Adjoint rep is self conjugate since

$$\text{ad}_\mathfrak{h}(E^\alpha) = \alpha E^\alpha \text{ and } \text{ad}_\mathfrak{h}(E^{-\alpha}) = -\alpha E^{-\alpha}$$

$$\Leftrightarrow E^\alpha = -E^{-\alpha}$$

$$0-0-0 \quad 0-0-0$$

* Diagrams w/ reflection symmetry: A_r, D_r, E_6 

- Conj. Seen as a reflection is simply an automorphism of the weight space: Each node of the Dynkin diagram corresponding to the fundamental weights

* In order to have the automorphism on A_r, B_r, E_6 we must have:

$$A_r : \lambda_i = \lambda_{r-i+1}$$

$$D_r : \lambda_r = \lambda_{r-1} \quad (r = 2m)$$

$$E_6 : \lambda_1 = \lambda_5 \text{ \& \& } \lambda_2 = \lambda_6$$

(Aside) Highest weight reps in Physics

$$\text{Cartan} : A_{ij} := \langle \alpha_i, \alpha_j^\vee \rangle$$

A generic weight λ has an expansion in $\alpha_i, \alpha_i^\vee$ or ω_i

- Fundamental weights: $\text{Span}\{\omega_i\} = \mathcal{H}^* \leftarrow \text{Cartan} - \text{subalg.}$

$$\langle \alpha_i, \alpha_j \rangle = A_{ij} |\alpha_i|^2 / 2$$

$$\langle \omega_i, \omega_j \rangle = F_{ij}$$

$$\langle \alpha_i^\vee, \alpha_j^\vee \rangle = \frac{2}{|\alpha_i|^2} A_{ij}$$

$$\lambda_i = \langle \lambda, \alpha_i^\vee \rangle = \sum_{i=1}^n K_i A_{ij}$$

$$K_i = \sum_{j=1}^n \lambda_j \frac{2}{|\alpha_i|^2} F_{ji}$$

$$\therefore (\hat{F}^{-1})_{ij} = \frac{2}{|\alpha_i|^2} A_{ij} = \langle \alpha_i^\vee, \alpha_j^\vee \rangle \quad (\text{Quadratic Form})$$

$$\Rightarrow \lambda = K^{(1)} \hat{A} \quad \text{or} \quad K^{(1)} = \lambda \hat{A}^{-1} \quad \lambda, K^{(1)} \text{ "row" vectors}$$

$$\text{ex: } \underline{3} \text{ of } SU(3): \quad \mathcal{B}(\underline{3}, \mathbb{C}) = \mathcal{SL}(\mathbb{C}^3, \mathbb{C}) \quad \text{---}$$

$$\underline{A}_2: \quad \underline{3} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = A_{ij}$$

$$\downarrow \alpha_1$$

$$\begin{bmatrix} -1 & 2 \end{bmatrix}$$

$$\downarrow -\alpha_2$$

$$\begin{bmatrix} 0 & -1 \end{bmatrix}$$

$$\underline{3}: [10], \quad \bar{\underline{3}}: [01], \quad [1,1]: \alpha_{ij} \quad \square$$

$$(\hat{x} \phi a)_i = x_{ia}^{(1)} \phi a_i + x_{ia}^{(2)} a_i \phi a$$

$$SU(n+1): \quad \underline{n+1} : \Delta_n \quad \underline{n+1} = [10 \dots 0]$$

$$[10 \dots 0]$$

↓

$$[-110 \dots]$$

↓

$$[0 \dots 0 -11]$$

↓

$$[00 \dots 0 -2]$$

$$\underline{n=2}: T_{[ij]}, \left(\frac{n+1}{2}\right)$$

• The highest weight for each comp. is $[10 \dots 0]$ however, it's 0 for j we take $[-110 \dots 0]$

$$\rho_{\text{succin}} T_{[ij]} = [10 \dots 0] \oplus [-11 \dots 0] = [010 \dots 0] \rightarrow \omega_2$$

$$\underline{n=3}: \rho_{\text{succin}} T_{[ijk]} = [10 \dots 0] \oplus [-11 \dots 0] \oplus [0-110 \dots] = [0010 \dots 0] \rightarrow \omega_3$$

$$\underline{n=n}: T_{[i_1 \dots i_n]} = [1 \dots 0] \oplus \dots \oplus [0 \dots -11] = [0 \dots 01] \rightarrow \omega_n$$

$$M=n+1: \bar{T}_{[i_1 \dots i_{n+1}]} = 0$$

$$\ast \sum U(n+1): \quad \begin{matrix} \bar{T}_{[i_1 i_2]} & \dots & \bar{T}_{[i_{n-1} i_n]} \\ 0-0-\dots-0-0 \end{matrix} \sim \begin{matrix} \square & \square & & \begin{matrix} \square \\ \square \\ \square \end{matrix} \\ 0-0-\dots-0-0 \end{matrix} \Bigg\}^n$$

ω_i correspond to rank- n anti-symm. tensors.

• This also applies to all but two fundamental weights in $B_n = SO(2n+1)$, $D_n = SO(2n)$

$$0-0-\alpha-\dots-0 \Rightarrow 0$$

$$0-0-\alpha-\dots-0 \Rightarrow 0$$

→ Note, for B_n we only have one while D_n has two. This is owed to the fact that $SO(2n)$ admits two different spinor reps & B_n only one.

We shall prove this below, however we can immediately note:

$$\underline{B_n}: \dim(\mathfrak{so}(2n+1)) = n$$

$$[1 \dots 0] T_i, [01 \dots 0] T_{[i,j]}, [001 \dots 0] T_{[i,j,k]}, [0 \dots 10] T_{[i \dots i, n-1]}$$

$$[0 \dots 1] \chi_\alpha$$

$$\underline{D_n}: [0 \dots 10] \chi^\alpha \quad [0 \dots 01] \chi_\alpha$$

$$[0 \dots 0100] T_{[i_1 \dots i_{n-2}]}$$

Symmetrization: $T_{[i_1 \dots i_m]}$ $m \geq 1$

For each index we are not restricted in choosing highest weights so naturally

$$[m0 \dots 0] : T_{[i_1 \dots i_m]}$$

* Also true for C_n , where $0-0 \dots 0 \leq 0$ although not quite for B_n & D_n .

Levi-civita, $\mathfrak{sl}(n+1)$ rep of $SL(n+1)$ & Conj. rep's.

$$e_{i_1 \dots i_n} = 1,$$

* $(\mathfrak{sl}(n+1))$ different ω_i -indices are needed to form a singlet $\frac{1}{n}$ in $SL(n+1)$

→ This give an alternate tensor rep for ω_n

The conjugate $\mathfrak{sl}(n+1)$ rep:

$$T^i := \varepsilon^{i_1 \dots i_n} T_{[i_1 \dots i_n]}, \quad \overline{\mathfrak{sl}(n+1)} = \omega_n \text{ Conj. rep of } SL(n+1)$$

* Given T^i_j of $SL(n+1)$ note $T^i \sim \varepsilon^{i_1 \dots i_n} T_{[i_1 \dots i_n]}$ while T_j is simply $[10 \dots 0]$.

$$(i) \text{ Scalars: } T_0 := T^i_i \quad (0)$$

$$(ii) \text{ Vectors: } \tilde{T}_i := T^j_i \quad \frac{\delta^j_i}{n} = \frac{1}{n} \quad \square$$

$$[U]^\dagger \text{Adjoint: } U^\dagger_{ij} := U_{ji} = (U^{-1})^k_k$$

* The pairwise contraction of defining $n+2$ and the conjugate $\overline{n+1}$ rep give a scalar (0). Alternatively, by duality of forms:

$$U^i V_i = \epsilon^{i_1 \dots i_{n+1} i_{n+2}} U_{[i_1 \dots i_{n+1}]} V_{i_{n+2}}$$

* Physically: This type of dual view occurs for symmetry transf. of complex quantum fields

$$\text{If: } \psi \mapsto \hat{U} \psi, \psi^\dagger \mapsto \psi^\dagger \hat{U}^\dagger, U U^\dagger = \underline{1}$$

Then, the bilinears

$$\psi^\dagger \psi \mapsto \psi^\dagger \psi \text{ (invariant scalar)}$$

$$\psi^\dagger \hat{M} \psi \mapsto \psi^\dagger [\hat{U}^\dagger \hat{M} \hat{U}] \psi \text{ (adjoint rep for } \text{Tr}(\hat{M}) = 0)$$

• If ψ_i [$\psi^\dagger_i$] transform in the $n+2$ [$\overline{n+1}$] of $SU(n+1)$ then M^i_j can be decomposed into linear combination of $SU(n+1)$ generators

- ▽ Alternate way to define the conjugate rep:

$$\psi_i \mapsto U_i^j \psi_j, U_i^j = [\exp(\hat{T}_A \theta^A)]_i^j$$

$$\psi^\dagger_i \mapsto (\psi^\dagger)^j (U^\dagger)^j_i, \hat{U}^\dagger = \exp(\hat{T}_A^\dagger (\theta^A)^\dagger)$$

• Suppose we have T_A s.t. we only span $\mathfrak{h} \subset \mathfrak{g}$, i.e. the Cartan sub-algebra. $\hat{T}_A \in \{\hat{H}_{\alpha_i}^{(n+1)}\} \Leftrightarrow (\hat{T}_A)^\dagger = \hat{T}_A$ & $\theta_A \in \mathbb{R}$
 $H_i^{(n+1)} = -H_i^{(n+1)}$

ex: Defining $n+2$ of $SU(n+1)$

$$|1\rangle := \begin{pmatrix} 1 \\ 0 \\ \vdots \end{pmatrix}, \Lambda^{(1)} = [1 \dots 0] \text{ (highest weight state)}$$

$$|2\rangle := \begin{pmatrix} 0 \\ 1 \\ \vdots \end{pmatrix}, \Lambda^{(2)} := [-1 \dots 0]$$

$$|n+1\rangle = \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix}, \Lambda^{(n+1)} := [0 \dots 0 -1] \text{ (lowest weight state)}$$

$\Rightarrow$ The weight vectors & weights $\in \overline{n+1}$

$$|\bar{1}\rangle := \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix} = \Lambda^{(2,1)} = [-1 \dots 0] \text{ (lowest weight state)}$$

$$\Lambda^{(n)} = \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix}, \quad -\Lambda^{(n+1)} = [0 \dots 1] \quad (\text{lowest weight state})$$

• Note: since the conj. rep is simply the negative of $n+1$ rep, $U_1 V^1 + U_2 V^2 + \dots$

each term cancels the weight!

$$\text{i.e. } \rho_{\text{SO}(n+1)}(U_1 V^1) = \rho(U) V + U \rho(V) \\ \Rightarrow [1 \dots 0] + [-1 \dots 0]^0$$

Def: A rep λ is self conjugate if its lowest weight is equal to minus its highest weight

• Otherwise, the rep is said to be complex and $\exists$ a conjugate rep λ^* which has equal weights to those in λ times -1 .

$$\hookrightarrow \Delta(2n+1) \quad B_n: \quad 0-0-0-0 \Rightarrow 0 \quad A_{ij} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix}$$

$$\text{ex: } n=3: \hookrightarrow \Delta(7) \quad 0-0 \Rightarrow 0$$

$[100]: \mathbb{Z}$, $\mathbb{Z}$ is a self conjugate rep of $\text{SO}(2)$

$$\downarrow -\alpha_1 \quad * [010] = [100] \oplus [-110] \quad \text{i.e. } T_{\Sigma_{ij}}$$

$$\downarrow -\alpha_2 \quad [0-12] \quad \bullet \text{ Note } T_{\Sigma_{ik}}: [100] \oplus [-110] \oplus [0-22]$$

$$\downarrow \alpha_3 \quad [0003] = [002]! \quad \text{NOT } [001]$$

$$\downarrow -\alpha_1 \quad [01-2]$$

$$\downarrow \alpha_2 \quad [1-10]$$

$$\downarrow -\alpha_1 \quad [-100]$$

Higher rank tensors:

$$T_{\Sigma_{i_1 i_2 i_3}} = \varepsilon_{i_1 i_2 i_3 \dots i_7} T^{[i_1 i_2 i_3 i_4 i_5 i_6 i_7]} \Leftrightarrow [002]$$

$$T_{\Sigma_{i_1 i_2}} = \varepsilon_{i_1 i_2 \dots i_7} T^{[i_1 i_2 \dots i_7]} \Leftrightarrow [010]$$

$$T_{i_1} = \varepsilon_{i_1 \dots i_7} T^{[i_1 \dots i_7]} \Leftrightarrow [100]$$

$$T = \varepsilon_{i_1 \dots i_7} T^{[i_1 \dots i_7]} \Leftrightarrow (0)$$

Q: What is $[001]$? $[001]$

$$\dim([001]) = 8 = 2^n$$

$$\downarrow -\alpha_3 \quad [01-2]$$

$$\downarrow -\alpha_2 \quad [1-11]$$

$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix}$$

$[002]$: fundamental spinor

$$\begin{array}{c}
 \begin{array}{cc}
 \begin{array}{c} -\alpha_1 \swarrow \\ [-1 \ 0 \ 1] \end{array} & \begin{array}{c} \searrow -\alpha_3 \\ [1 \ 0 \ -1] \end{array} \\
 \begin{array}{c} \swarrow -\alpha_3 \\ [-1 \ 1 \ -1] \end{array} & \begin{array}{c} \nwarrow -\alpha_1 \\ [-1 \ 1 \ -1] \end{array} \\
 \downarrow -\alpha_2 \\
 [0 \ -1 \ 1] \\
 \downarrow -\alpha_3 \\
 [0 \ 0 \ -1]
 \end{array}
 \end{array}$$

* Note, for the fundamental vector rep of B_3 , the weights are all orthogonal $\langle \lambda^{(a)}, \lambda^{(b)} \rangle = \delta^{ab}$

→ The spinor rep here (i.e. in odd dimensions) cannot be obtained by tensoring together reps.